JEE MAIN · MATHEMATICS · CHAPTER-WISE PYQS

Trigonometric Ratio and Identites

JEE Main Mathematics · Previous Year Questions with Answer Key

68	65	2004 – 2026	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

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1 **Solve**
Print it, or work through it on screen.

2 **Check**
The answer key is on the last pages.

MCQ (Single Correct Answer) 59 questions

Q1. JEE Main 2026 (Online) 5th April Morning Shift

Let $\tan A, \tan B$, where $A, B \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, be the roots of the quadratic equation $x^2 - 2x - 5 = 0$. Then $20\sin^2\left(\frac{A+B}{2}\right)$ is equal to:

(A) $10 + \sqrt{10}$ (B) $10 - 2\sqrt{10}$ (C) 10-3 $\sqrt{10}$ (D) $10 - \sqrt{10}$

Q2. JEE Main 2026 (Online) 2nd April Evening Shift

Let $P = \{\theta \in [0, 4\pi] : \tan^2\theta \neq 1\}$ and $S = \{a \in \mathbb{Z} : 2(\cos^8\theta - \sin^8\theta)\sec 2\theta = a^2, \theta \in P\}$. Then $n(S)$ is:

(A) 0 (B) 1 (C) 2 (D) 3

Q3. JEE Main 2026 (Online) 2nd April Morning Shift

If $\sin\left(\frac{\pi}{18}\right)\sin\left(\frac{5\pi}{18}\right)\sin\left(\frac{7\pi}{18}\right) = K$, then the value of $\sin\left(\frac{10K\pi}{3}\right)$ is:

(A) $\frac{\sqrt{3}+1}{2\sqrt{2}}$ (B) $\frac{\sqrt{3}-1}{\sqrt{2}}$ (C) $\frac{\sqrt{3}}{2}$ (D) $\frac{1}{2}$

Q4. JEE Main 2026 (Online) 28th January Morning Shift

If $\frac{\tan(\mathrm{A}-\mathrm{B})}{\tan \mathrm{A}} + \frac{\sin^2 \mathrm{C}}{\sin^2 \mathrm{A}} = 1,\ \mathrm{A}, \mathrm{B}, \mathrm{C} \in \left(0, \frac{\pi}{2}\right)$, then

(A) $\tan \mathrm{A}, \tan \mathrm{C}, \tan \mathrm{B}$ are in A.P. (B) $\tan \mathrm{A}, \tan \mathrm{C}, \tan \mathrm{B}$ are in G.P.

(C) $\tan A, \tan B, \tan C$ are in G.P. (D) $\tan A, \tan B, \tan C$ are in A.P.

Q5. JEE Main 2026 (Online) 24th January Morning Shift

The value of $\frac{\sqrt{3}\,\mathrm{cosec}\,20^\circ - \sec 20^\circ}{\cos 20^\circ \cos 40^\circ \cos 60^\circ \cos 80^\circ}$ is equal to

(A) 32 (B) 64 (C) 12 (D) 16

Q6. JEE Main 2026 (Online) 24th January Morning Shift

If $\cot x = \frac{5}{12}$ for some $x \in \left(\pi, \frac{3\pi}{2}\right)$, then $\sin 7x\left(\cos\frac{13x}{2} + \sin\frac{13x}{2}\right) + \cos 7x\left(\cos\frac{13x}{2} - \sin\frac{13x}{2}\right)$ is equal to

(A) $\frac{5}{\sqrt{13}}$ (B) $\frac{6}{\sqrt{26}}$ (C) $\frac{4}{\sqrt{26}}$ (D) $\frac{1}{\sqrt{13}}$

Q7. JEE Main 2026 (Online) 23rd January Evening Shift

Let $\frac{\pi}{2} < \theta < \pi$ and $\cot\theta = -\frac{1}{2\sqrt{2}}$. Then the value of

$\sin\left(\frac{15\theta}{2}\right)(\cos 8\theta + \sin 8\theta) + \cos\left(\frac{15\theta}{2}\right)(\cos 8\theta - \sin 8\theta)$

is equal to :

(A) $\frac{\sqrt{2}-1}{\sqrt{3}}$ (B) $\frac{\sqrt{2}}{\sqrt{3}}$ (C) $\frac{1-\sqrt{2}}{\sqrt{3}}$ (D) $-\frac{\sqrt{2}}{\sqrt{3}}$

Q8. JEE Main 2026 (Online) 21st January Morning Shift

The value of $\operatorname{cosec} 10^{\circ}-\sqrt{3} \sec 10^{\circ}$ is equal to :

(A) 2 (B) 6 (C) 8 (D) 4

Q9. JEE Main 2025 (Online) 7th April Morning Shift

If for $\theta \in\left[-\frac{\pi}{3}, 0\right]$, the points $(x, y)=\left(3 \tan \left(\theta+\frac{\pi}{3}\right), 2 \tan \left(\theta+\frac{\pi}{6}\right)\right)$ lie on $xy+\alpha x+\beta y+\gamma=0$, then $\alpha^2+\beta^2+\gamma^2$ is equal to :

(A) 75 (B) 96 (C) 80 (D) 72

Q10. JEE Main 2025 (Online) 4th April Morning Shift

If $10 \sin^4 \theta+15 \cos^4 \theta=6$, then the value of $\frac{27 \operatorname{cosec}^6 \theta+8 \sec^6 \theta}{16 \sec^8 \theta}$ is

(A) $\frac{2}{5}$ (B) $\frac{3}{5}$ (C) $\frac{1}{5}$ (D) $\frac{3}{4}$

Q11. JEE Main 2025 (Online) 29th January Evening Shift

If $\sin x+\sin^2 x=1, x \in\left(0, \frac{\pi}{2}\right)$, then

$\left(\cos^{12} x+\tan^{12} x\right)+3\left(\cos^{10} x+\tan^{10} x+\cos^8 x+\tan^8 x\right)+\left(\cos^6 x+\tan^6 x\right)$ is equal to:

(A) 3 (B) 4 (C) 2 (D) 1

Q12. JEE Main 2025 (Online) 28th January Evening Shift

If $\sum_{r=1}^{13}\left\{\frac{1}{\sin \left(\frac{\pi}{4}+(r-1) \frac{\pi}{6}\right) \sin \left(\frac{\pi}{4}+\frac{r \pi}{6}\right)}\right\}=a \sqrt{3}+b, a, b \in Z$, then a^2+b^2 is equal to :

(A) 10 (B) 4 (C) 8 (D) 2

Q13. JEE Main 2025 (Online) 23rd January Evening Shift

Let the range of the function $f(x)=6+16 \cos x \cdot \cos \left(\frac{\pi}{3}-x\right) \cdot \cos \left(\frac{\pi}{3}+x\right) \cdot \sin 3 x \cdot \cos 6 x, x \in \mathbf{R}$ be $[\alpha, \beta]$. Then the distance of the point (α, β) from the line $3x+4y+12=0$ is :

(A) 11 (B) 10 (C) 8 (D) 9

Q14. JEE Main 2025 (Online) 23rd January Morning Shift

The value of $(\sin 70^{\circ})\left(\cot 10^{\circ} \cot 70^{\circ}-1\right)$ is

(A) 0 (B) 2/3 (C) 1 (D) 3/2

Q15. JEE Main 2024 (Online) 8th April Evening Shift

If the value of $\frac{3\cos 36^\circ+5\sin 18^\circ}{5\cos 36^\circ-3\sin 18^\circ}$ is $\frac{a\sqrt{5}-b}{c}$, where a, b, c are natural numbers and $\gcd(a, c) = 1$, then $a + b + c$ is equal to :

(A) 54 (B) 52 (C) 50 (D) 40

Q16. JEE Main 2024 (Online) 8th April Morning Shift

If $\sin x = -\frac{3}{5}$, where $\pi < x < \frac{3\pi}{2}$, then $80\left(\tan^2 x - \cos x\right)$ is equal to

(A) 109 (B) 108 (C) 19 (D) 18

Q17. JEE Main 2024 (Online) 1st February Morning Shift

If $\tan \mathrm{A} = \frac{1}{\sqrt{x(x^2+x+1)}}$, $\tan \mathrm{B} = \frac{\sqrt{x}}{\sqrt{x^2+x+1}}$ and

$\tan \mathrm{C} = \left(x^{-3} + x^{-2} + x^{-1}\right)^{1/2}$, $0 < \mathrm{A}, \mathrm{B}, \mathrm{C} < \frac{\pi}{2}$, then $\mathrm{A} + \mathrm{B}$ is equal to :

(A) C (B) $\pi - C$ (C) $2\pi - C$ (D) $\frac{\pi}{2} - \mathrm{C}$

Q18. JEE Main 2024 (Online) 31st January Evening Shift

The number of solutions, of the equation $e^{\sin x} - 2e^{-\sin x} = 2$, is :

(A) 0 (B) 1 (C) 2 (D) more than 2

Q19. JEE Main 2024 (Online) 30th January Evening Shift

For $\alpha, \beta \in (0, \pi/2)$, let $3\sin(\alpha + \beta) = 2\sin(\alpha - \beta)$ and a real number k be such that $\tan\alpha = k\tan\beta$. Then, the value of k is equal to

(A) 5 (B) $-2/3$ (C) -5 (D) 2/3

Q20. JEE Main 2023 (Online) 10th April Morning Shift

$96\cos\frac{\pi}{33}\cos\frac{2\pi}{33}\cos\frac{4\pi}{33}\cos\frac{8\pi}{33}\cos\frac{16\pi}{33}$ is equal to :

(A) 4 (B) 2 (C) 1 (D) 3

Q21. JEE Main 2023 (Online) 8th April Evening Shift

The value of $36\left(4\cos^2 9^\circ - 1\right)\left(4\cos^2 27^\circ - 1\right)\left(4\cos^2 81^\circ - 1\right)\left(4\cos^2 243^\circ - 1\right)$ is :

(A) 18 (B) 36 (C) 54 (D) 27

Q22. JEE Main 2023 (Online) 30th January Morning Shift

If $\tan 15^\circ + \frac{1}{\tan 75^\circ} + \frac{1}{\tan 105^\circ} + \tan 195^\circ = 2a$, then the value of $\left(a + \frac{1}{a}\right)$ is :

(A) $5 - \frac{3}{2}\sqrt{3}$ (B) $4 - 2\sqrt{3}$ (C) 2 (D) 4

Q23. JEE Main 2023 (Online) 29th January Evening Shift

The set of all values of λ for which the equation $\cos^2 2x - 2\sin^4 x - 2\cos^2 x = \lambda$ has a real solution x, is :

(A) $[-2, -1]$ (B) $\left[-\frac{3}{2}, -1\right]$

(C) $\left[-2, -\frac{3}{2}\right]$ (D) $\left[-1, -\frac{1}{2}\right]$

Q24. JEE Main 2023 (Online) 29th January Morning Shift

Let $f(\theta) = 3\left(\sin^4\left(\frac{3\pi}{2} - \theta\right) + \sin^4(3\pi + \theta)\right) - 2(1 - \sin^2 2\theta)$ and $S = \left\{\theta \in [0, \pi] : f'(\theta) = -\frac{\sqrt{3}}{2}\right\}$. If $4\beta = \sum_{\theta \in S} \theta$, then $f(\beta)$ is equal to

(A) $\frac{9}{8}$ (B) $\frac{3}{2}$ (C) $\frac{5}{4}$ (D) $\frac{11}{8}$

Q25. JEE Main 2022 (Online) 25th July Evening Shift

$2\sin\left(\frac{\pi}{22}\right)\sin\left(\frac{3\pi}{22}\right)\sin\left(\frac{5\pi}{22}\right)\sin\left(\frac{7\pi}{22}\right)\sin\left(\frac{9\pi}{22}\right)$ is equal to :

(A) $\frac{3}{16}$ (B) $\frac{1}{16}$ (C) $\frac{1}{32}$ (D) $\frac{9}{32}$

Q26. JEE Main 2022 (Online) 28th June Evening Shift

If $\cot\alpha$ = 1 and $\sec\beta = -\frac{5}{3}$, where $\pi < \alpha < \frac{3\pi}{2}$ and $\frac{\pi}{2} < \beta < \pi$, then the value of $\tan(\alpha + \beta)$ and the quadrant in which $\alpha + \beta$ lies, respectively are :

(A) $-\frac{1}{7}$ and IVth quadrant (B) 7 and Ist quadrant

(C) -7 and IVth quadrant (D) $\frac{1}{7}$ and Ist quadrant

Q27. JEE Main 2022 (Online) 27th June Evening Shift

$\alpha = \sin 36^\circ$ is a root of which of the following equation?

(A) $16x^4 - 10x^2 - 5 = 0$ (B) $16x^4 + 20x^2 - 5 = 0$

(C) $16x^4 - 20x^2 + 5 = 0$ (D) $4x^4 - 10x^2 + 5 = 0$

Q28. JEE Main 2022 (Online) 27th June Morning Shift

The value of $\cos\left(\frac{2\pi}{7}\right) + \cos\left(\frac{4\pi}{7}\right) + \cos\left(\frac{6\pi}{7}\right)$ is equal to :

(A) -1 (B) $-\frac{1}{2}$ (C) $-\frac{1}{3}$ (D) $-\frac{1}{4}$

Q29. JEE Main 2022 (Online) 26th June Evening Shift

$16\sin(20^\circ)\sin(40^\circ)\sin(80^\circ)$ is equal to :

(A) $\sqrt{3}$ (B) $2\sqrt{3}$ (C) 3 (D) $4\sqrt{3}$

Q30. JEE Main 2022 (Online) 25th June Evening Shift

The value of $2\sin(12^\circ) - \sin(72^\circ)$ is :

(A) $\frac{\sqrt{5}(1-\sqrt{3})}{4}$ (B) $\frac{1-\sqrt{5}}{8}$

(C) $\frac{\sqrt{3}(1-\sqrt{5})}{2}$ (D) $\frac{\sqrt{3}(1-\sqrt{5})}{4}$

Q31. JEE Main 2021 (Online) 26th August Evening Shift

The value of

$2\sin\left(\frac{\pi}{8}\right)\sin\left(\frac{2\pi}{8}\right)\sin\left(\frac{3\pi}{8}\right)\sin\left(\frac{5\pi}{8}\right)\sin\left(\frac{6\pi}{8}\right)\sin\left(\frac{7\pi}{8}\right)$ is :

(A) $\frac{1}{4\sqrt{2}}$ (B) $\frac{1}{4}$ (C) $\frac{1}{8}$ (D) $\frac{1}{8\sqrt{2}}$

Q32. JEE Main 2021 (Online) 27th July Evening Shift

If $\tan\left(\frac{\pi}{9}\right), x, \tan\left(\frac{7\pi}{18}\right)$ are in arithmetic progression and $\tan\left(\frac{\pi}{9}\right), y, \tan\left(\frac{5\pi}{18}\right)$ are also in arithmetic progression, then $|x-2y|$ is equal to :

(A) 4 (B) 3 (C) 0 (D) 1

Q33. JEE Main 2021 (Online) 27th July Morning Shift

If $\sin\theta + \cos\theta = \frac{1}{2}$, then 16(sin(2$\theta$) + cos(4$\theta$) + sin(6$\theta$)) is equal to :

(A) 23 (B) -27 (C) -23 (D) 27

Q34. JEE Main 2021 (Online) 25th July Evening Shift

The value of $\cot\frac{\pi}{24}$ is :

(A) $\sqrt{2}+\sqrt{3}+2-\sqrt{6}$ (B) $\sqrt{2}+\sqrt{3}+2+\sqrt{6}$

(C) $\sqrt{2}-\sqrt{3}-2+\sqrt{6}$ (D) $3\sqrt{2}-\sqrt{3}-\sqrt{6}$

Q35. JEE Main 2021 (Online) 18th March Evening Shift

If $15\sin^4\alpha + 10\cos^4\alpha = 6$, for some $\alpha \in R$, then the value of

$27\sec^6\alpha + 8\text{cosec}^6\alpha$ is equal to :

(A) 500 (B) 400 (C) 250 (D) 350

Q36. JEE Main 2021 (Online) 16th March Morning Shift

If for $x \in \left(0, \frac{\pi}{2}\right)$, $\log_{10}\sin x + \log_{10}\cos x = -1$ and $\log_{10}(\sin x + \cos x) = \frac{1}{2}(\log_{10} n - 1)$, $n > 0$, then the value of n is equal to :

(A) 16 (B) 9 (C) 12 (D) 20

Q37. JEE Main 2021 (Online) 25th February Evening Shift

If $0 < x, y < \pi$ and $\cos x + \cos y - \cos(x + y) = \frac{3}{2}$, then $\sin x + \cos y$ is equal to :

(A) $\frac{1+\sqrt{3}}{2}$ (B) $\frac{1}{2}$ (C) $\frac{\sqrt{3}}{2}$ (D) $\frac{1-\sqrt{3}}{2}$

Q38. JEE Main 2021 (Online) 24th February Morning Shift

If $e^{(\cos^2 x + \cos^4 x + \cos^6 x + \ldots\infty)\log_e 2}$ satisfies the equation $t^2 - 9t + 8 = 0$, then the value of $\frac{2\sin x}{\sin x + \sqrt{3}\cos x}\left(0 < x < \frac{\pi}{2}\right)$ is :

(A) $\sqrt{3}$ (B) $\frac{3}{2}$ (C) $2\sqrt{3}$ (D) $\frac{1}{2}$

Q39. JEE Main 2020 (Online) 5th September Evening Slot

If $L = \sin^2\left(\frac{\pi}{16}\right) - \sin^2\left(\frac{\pi}{8}\right)$ and
$M = \cos^2\left(\frac{\pi}{16}\right) - \sin^2\left(\frac{\pi}{8}\right)$, then :

(A) $L = -\frac{1}{2\sqrt{2}} + \frac{1}{2}\cos\frac{\pi}{8}$ (B) $M = \frac{1}{2\sqrt{2}} + \frac{1}{2}\cos\frac{\pi}{8}$

(C) $M = \frac{1}{4\sqrt{2}} + \frac{1}{4}\cos\frac{\pi}{8}$ (D) $L = \frac{1}{4\sqrt{2}} - \frac{1}{4}\cos\frac{\pi}{8}$

Q40. JEE Main 2020 (Online) 2nd September Evening Slot

If the equation $\cos^4\theta + \sin^4\theta + \lambda = 0$ has real solutions for θ, then λ lies in the interval :

(A) $\left[-\frac{3}{2}, -\frac{5}{4}\right]$ (B) $\left(-\frac{1}{2}, -\frac{1}{4}\right]$

(C) $\left(-\frac{5}{4}, -1\right]$ (D) $\left[-1, -\frac{1}{2}\right]$

Q41. JEE Main 2020 (Online) 9th January Evening Slot

If $x = \sum_{n=0}^{\infty}(-1)^n \tan^{2n}\theta$ and $y = \sum_{n=0}^{\infty}\cos^{2n}\theta$

for $0 < \theta < \frac{\pi}{4}$, then :

(A) x(1 + y) = 1 (B) y(1 − x) = 1 (C) y(1 + x) = 1 (D) x(1 − y) = 1

Q42. JEE Main 2020 (Online) 9th January Morning Slot

The value of
$\cos^3\left(\frac{\pi}{8}\right)\cos\left(\frac{3\pi}{8}\right)+\sin^3\left(\frac{\pi}{8}\right)\sin\left(\frac{3\pi}{8}\right)$
is :

(A) $\frac{1}{\sqrt{2}}$ (B) $\frac{1}{2}$ (C) $\frac{1}{4}$ (D) $\frac{1}{2\sqrt{2}}$

Q43. JEE Main 2019 (Online) 12th April Morning Slot

The equation y = sinx sin (x + 2) – $\sin^2$ (x + 1) represents a straight line lying in :

(A) first, second and fourth quadrants (B) first, third and fourth quadrants
(C) second and third quadrants only (D) third and fourth quadrants only

Q44. JEE Main 2019 (Online) 9th April Evening Slot

The value of sin 10° sin30° sin50° sin70° is :-

(A) $\frac{1}{36}$ (B) $\frac{1}{16}$ (C) $\frac{1}{32}$ (D) $\frac{1}{18}$

Q45. JEE Main 2019 (Online) 9th April Morning Slot

The value of $\cos^2 10° - \cos 10° \cos 50° + \cos^2 50°$ is

(A) $\frac{3}{2} + \cos 20^o$ (B) $\frac{3}{4}$ (C) $\frac{3}{2}(1 + \cos 20^o)$ (D) $\frac{3}{2}$

Q46. JEE Main 2019 (Online) 8th April Morning Slot

If $\cos(\alpha + \beta) = 3/5$, $\sin(\alpha - \beta) = 5/13$ and $0 < \alpha, \beta < \frac{\pi}{4}$, then $\tan(2\alpha)$ is equal to :

(A) 21/16 (B) 63/52 (C) 33/52 (D) 63/16

Q47. JEE Main 2019 (Online) 12th January Morning Slot

The maximum value of $3\cos\theta + 5\sin\left(\theta - \frac{\pi}{6}\right)$ for any real value of θ is :

(A) $\sqrt{34}$ (B) $\sqrt{31}$ (C) $\sqrt{19}$ (D) $\frac{\sqrt{79}}{2}$

Q48. JEE Main 2019 (Online) 10th January Evening Slot

The value of $\cos\frac{\pi}{2^2}.\cos\frac{\pi}{2^3}.....\cos\frac{\pi}{2^{10}}.\sin\frac{\pi}{2^{10}}$ is -

(A) $\frac{1}{256}$ (B) $\frac{1}{2}$ (C) $\frac{1}{1024}$ (D) $\frac{1}{512}$

Q49. JEE Main 2019 (Online) 9th January Morning Slot

For any $\theta \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$, the expression

$3(\cos\theta - \sin\theta)^4 + 6(\sin\theta + \cos\theta)^2 + 4\sin^6\theta$

equals :

(A) $13 - 4\cos^2\theta + 6\sin^2\theta\cos^2\theta$

(B) $13 - 4\cos^6\theta$

(C) $13 - 4\cos^2\theta + 6\cos^4\theta$

(D) $13 - 4\cos^4\theta + 2\sin^2\theta\cos^2\theta$

Q50. JEE Main 2017 (Offline)

If $5\left(\tan^2 x - \cos^2 x\right) = 2\cos 2x + 9$,

then the value of $\cos 4x$ is :

(A) $\frac{1}{3}$ (B) $\frac{2}{9}$ (C) $-\frac{7}{9}$ (D) $-\frac{3}{5}$

Q51. JEE Main 2016 (Online) 9th April Morning Slot

If m and M are the minimum and the maximum values of

$4 + \frac{1}{2}\sin^2 2x - 2\cos^4 x$, $x \in \mathbf{R}$, then $M - m$ is equal to :

(A) $\frac{15}{4}$ (B) $\frac{9}{4}$ (C) $\frac{7}{4}$ (D) $\frac{1}{4}$

Q52. JEE Main 2014 (Offline)

Let $f_k(x) = \frac{1}{k}\left(\sin^k x + \cos^k x\right)$ where $x \in R$ and $k \geq 1$.
Then $f_4(x) - f_6(x)$ equals :

(A) $\frac{1}{4}$ (B) $\frac{1}{12}$ (C) $\frac{1}{6}$ (D) $\frac{1}{3}$

Q53. JEE Main 2013 (Offline)

The expression $\frac{\tan A}{1-\cot A} + \frac{\cot A}{1-\tan A}$ can be written as:

(A) $\sin A \cos A + 1$

(B) $\sec A \cos ecA + 1$

(C) $\tan A + \cot A$

(D) $\sec A + \cos ecA$

Q54. AIEEE 2011

If $A = \sin^2 x + \cos^4 x$, then for all real x:

(A) $\frac{13}{16} \leq A \leq 1$

(B) $1 \leq A \leq 2$

(C) $\frac{3}{4} \leq A \leq \frac{13}{16}$

(D) $\frac{3}{4} \leq A \leq 1$

Q55. AIEEE 2010

Let $\cos(\alpha+\beta)=\frac{4}{5}$ and $\sin(\alpha-\beta)=\frac{5}{13}$, where $0\le\alpha,\beta\le\frac{\pi}{4}$.
Then $tan\,2\alpha$ =

(A) $\frac{56}{33}$ (B) $\frac{19}{12}$ (C) $\frac{20}{7}$ (D) $\frac{25}{16}$

Q56. AIEEE 2009

Let **A** and **B** denote the statements
A: $\cos\alpha+\cos\beta+\cos\gamma=0$

B: $\sin\alpha+\sin\beta+\sin\gamma=0$

If $\cos(\beta-\gamma)+\cos(\gamma-\alpha)+\cos(\alpha-\beta)=-\frac{3}{2}$, then:

(A) **A** is false and **B** is true (B) both **A** and **B** are true
(C) both **A** and **B** are false (D) **A** is true and **B** is false

Q57. AIEEE 2006

If $0<x<\pi$ and $\cos x+\sin x=\frac{1}{2}$, then $\tan x$ is :

(A) $\frac{\left(1-\sqrt{7}\right)}{4}$ (B) $\frac{\left(4-\sqrt{7}\right)}{3}$
(C) $-\frac{\left(4+\sqrt{7}\right)}{3}$ (D) $\frac{\left(1+\sqrt{7}\right)}{4}$

Q58. AIEEE 2004

Let α,β be such that $\pi<\alpha-\beta<3\pi$.
If $\sin\alpha+\sin\beta=-\frac{21}{65}$ and $\cos\alpha+\cos\beta=-\frac{27}{65}$ then the value of $\cos\frac{\alpha-\beta}{2}$:

(A) $\frac{-6}{65}$ (B) $\frac{3}{\sqrt{130}}$ (C) $\frac{6}{65}$ (D) $-\frac{3}{\sqrt{130}}$

Q59. AIEEE 2004

If $u=\sqrt{a^2\cos^2\theta+b^2\sin^2\theta}+\sqrt{a^2\sin^2\theta+b^2\cos^2\theta}$

then the difference between the maximum and minimum values of u^2 is given by :

(A) $(a-b)^2$ (B) $2\sqrt{a^2+b^2}$ (C) $(a+b)^2$ (D) $2\left(a^2+b^2\right)$

Numerical 9 questions

Q60. JEE Main 2026 (Online) 4th April Morning Shift

If $\mathrm{A}=\frac{\sin3^\circ}{\cos9^\circ}+\frac{\sin9^\circ}{\cos27^\circ}+\frac{\sin27^\circ}{\cos81^\circ}$ and $\mathrm{B}=\tan81^\circ-\tan3^\circ$, then $\frac{\mathrm{B}}{\mathrm{A}}$ is equal to

____ .

Q61. JEE Main 2026 (Online) 4th April Morning Shift

Let $\overrightarrow{a_k} = (\tan\theta_k)\hat{i} + \hat{j}$ and $\overrightarrow{b_k} = \hat{i} - (\cot\theta_k)\hat{j}$, where $\theta_k = \frac{2^{k-1}\pi}{2^n+1}$, for some $n \in \mathbb{N}, n > 5$. Then the value of $\frac{\sum_{k=1}^{n}\left|\overrightarrow{a_k}\right|^2}{\sum_{k=1}^{n}\left|\overrightarrow{b_k}\right|^2}$ is

Q62. JEE Main 2026 (Online) 22nd January Evening Shift

Let $\cos(\alpha+\beta) = -\frac{1}{10}$ and $\sin(\alpha-\beta) = \frac{3}{8}$, where $0 < \alpha < \frac{\pi}{3}$ and $0 < \beta < \frac{\pi}{4}$. If $\tan 2\alpha = \frac{3(1-r\sqrt{5})}{\sqrt{11}(s+\sqrt{5})}, r, s \in N$, then $r+s$ is equal to ____ .

Q63. JEE Main 2026 (Online) 22nd January Morning Shift

If $\frac{\cos^2 48^\circ - \sin^2 12^\circ}{\sin^2 24^\circ - \sin^2 6^\circ} = \frac{\alpha+\beta\sqrt{5}}{2}$, where $\alpha, \beta \in \mathbb{N}$, then $\alpha+\beta$ is equal to ____

Q64. JEE Main 2024 (Online) 27th January Morning Shift

Let the set of all $a \in \mathbf{R}$ such that the equation $\cos 2x + a\sin x = 2a - 7$ has a solution be $[p, q]$ and $r = \tan 9^\circ - \tan 27^\circ - \frac{1}{\cot 63^\circ} + \tan 81^\circ$, then pqr is equal to _____.

Q65. JEE Main 2023 (Online) 6th April Evening Shift

The value of $\tan 9^\circ - \tan 27^\circ - \tan 63^\circ + \tan 81^\circ$ is _____.

Q66. JEE Main 2022 (Online) 26th June Morning Shift

If $\sin^2(10^\circ)\sin(20^\circ)\sin(40^\circ)\sin(50^\circ)\sin(70^\circ) = \alpha - \frac{1}{16}\sin(10^\circ)$, then $16 + \alpha^{-1}$ is equal to _____.

Q67. JEE Main 2021 (Online) 26th February Morning Shift

The number of integral values of 'k' for which the equation $3\sin x + 4\cos x = k + 1$ has a solution, k∈R is _____.

Q68. JEE Main 2020 (Online) 8th January Evening Slot

If $\frac{\sqrt{2}\sin\alpha}{\sqrt{1+\cos 2\alpha}} = \frac{1}{7}$ and $\sqrt{\frac{1-\cos 2\beta}{2}} = \frac{1}{\sqrt{10}}$

$\alpha, \beta \in \left(0, \frac{\pi}{2}\right)$ then tan(α + 2β) is equal to _____.

Answer Key

MCQ (Single Correct Answer)

1. **C**	2. **A**	3. **A**	4. **B**	5. **B**
6. **D**	7. **C**	8. **D**	9. **A**	10. **A**
11. **C**	12. **C**	13. **A**	14. **C**	15. **B**
16. **A**	17. **A**	18. **A**	19. **C**	20. **D**
21. **B**	22. **D**	23. **B**	24. **C**	25. **B**
26. **A**	27. **C**	28. **B**	29. **B**	30. **D**
31. **C**	32. **C**	33. **C**	34. **B**	35. **C**
36. **C**	37. **A**	38. **D**	39. **B**	40. **D**
41. **B**	42. **D**	43. **D**	44. **B**	45. **B**
46. **D**	47. **C**	48. **D**	49. **B**	50. **C**
51. **B**	52. **B**	53. **B**	54. **D**	55. **A**
56. **B**	57. **C**	58. **D**	59. **A**	

Numerical

60. **2**	61. **3**	62. **20**	63. **4**	64. **48**
65. **4**	66. **80**	67. **11**	68. **1**	